Lecture 13

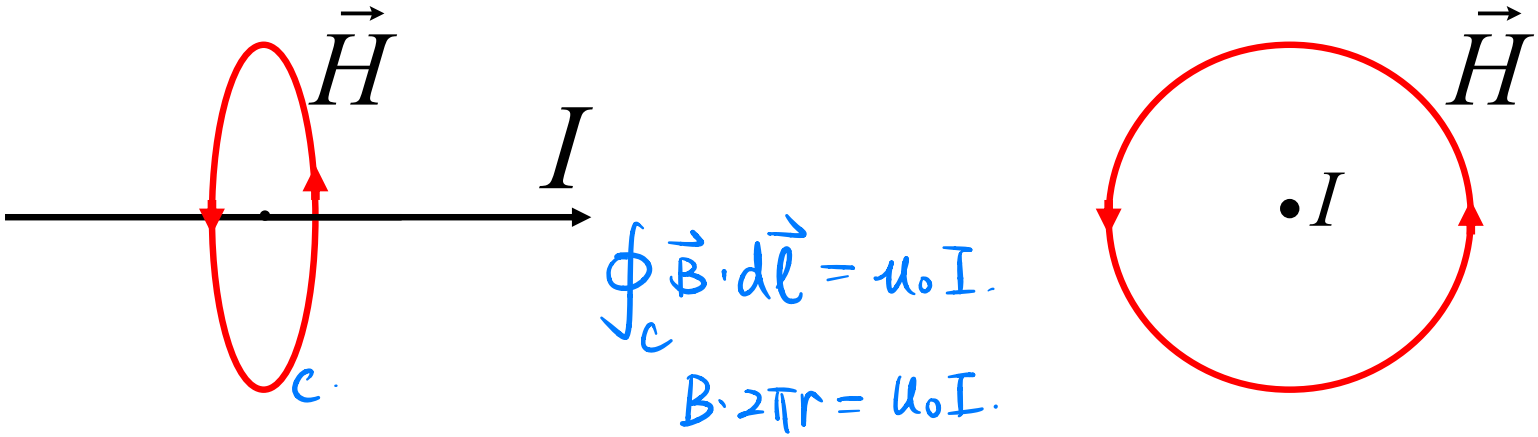
Ampere's Law

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"Static" Ampere's Law

Using the Biot-Savart Law, we solved for the **B** field around a straight wire and expressed it as **H**:



$$\oint_C \vec{B}\cdot d\vec{\ell} = \mu_0 I.$$

$$B\cdot 2\pi r = \mu_0 I.$$

$$\vec{B} = \frac{\mu_0 I}{2\pi r}\hat{a}_\phi$$

Magnetic flux density (Wb/m^2)

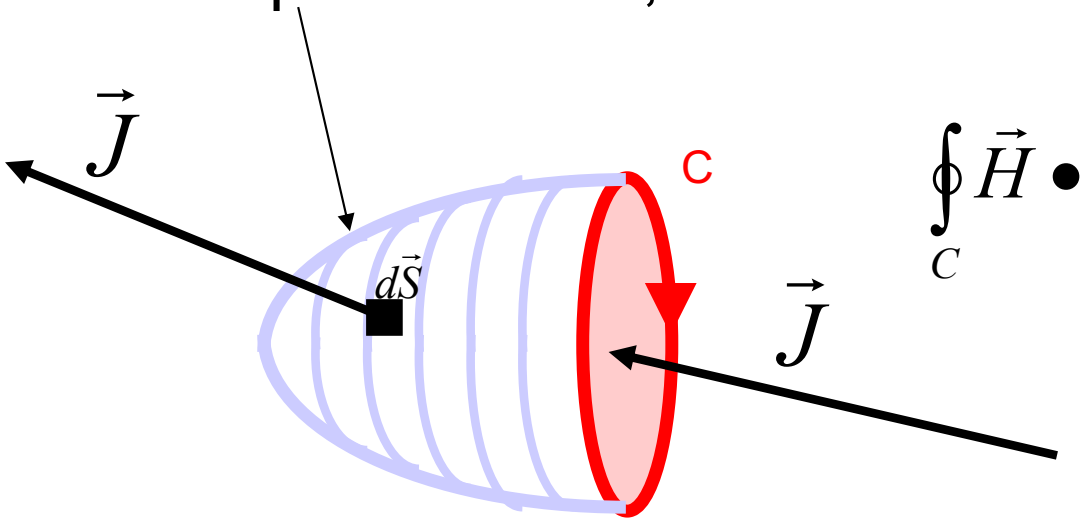
[T.]

$$\vec{H} = \frac{I}{2\pi r}\hat{a}_\phi = \frac{\vec{B}}{\mu_0}$$

Magnetic field (A/m)

Static Ampere's Law

Open Surface, where the Path forms a "Mouth"

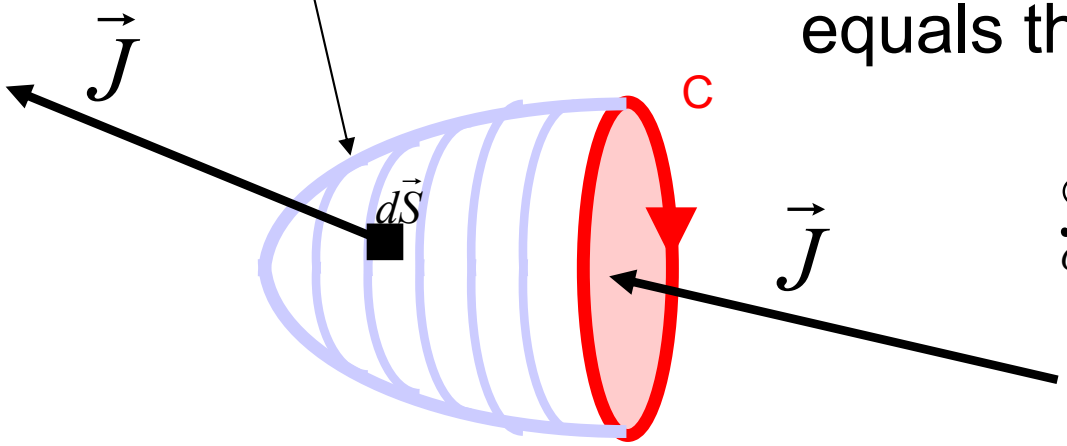


$$\oint_C \vec{H} \bullet d\vec{l} = I_{enclosed} = \iint_S \vec{J} \bullet d\vec{S}$$

In general, all the current entering the "mouth" will end up passing out of any surface that is formed around the mouth

Ampere's Law Physical Meaning

$$\oint_C \vec{H} \bullet d\vec{l} = I_{enclosed}$$



Taking the integral of **H** around a closed path equals the enclosed current.

$$\oint_C \vec{H} \bullet d\vec{l} = Magneto\,Motive\,Force\,(MMF)$$

$[\frac{A}{m}] \cdot [m]$ → $[A]$

→ like current source

similar to:

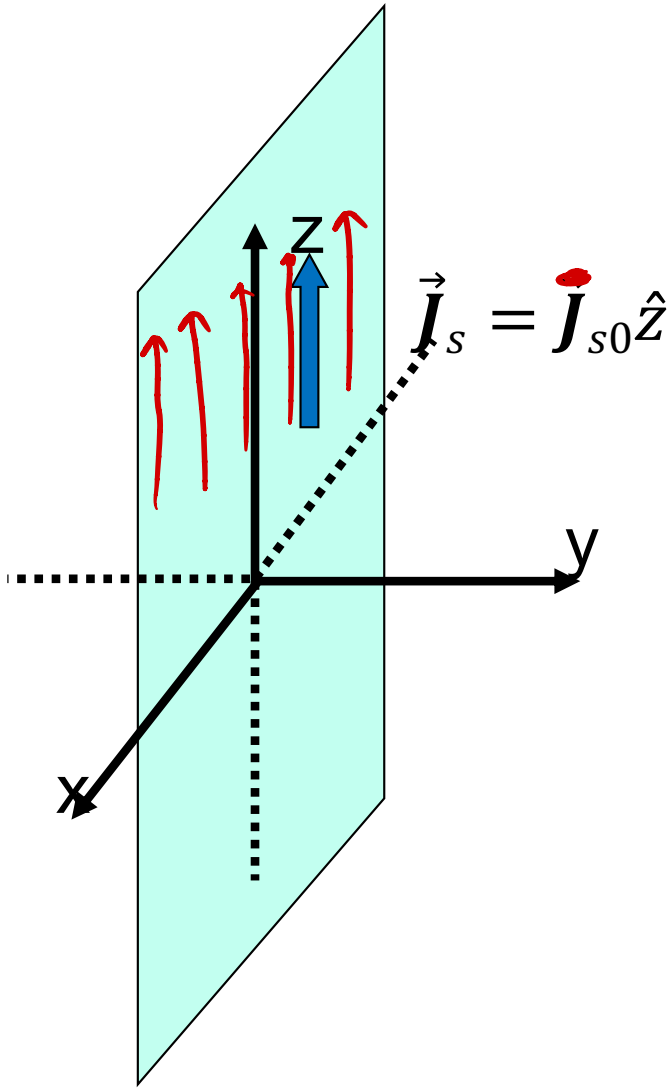
$$[V] = \oint_C \vec{E} \bullet d\vec{l} = Electro\,Motive\,Force\,(EMF)$$

→ like voltage source

Example: Infinite Plane Sheet of Current

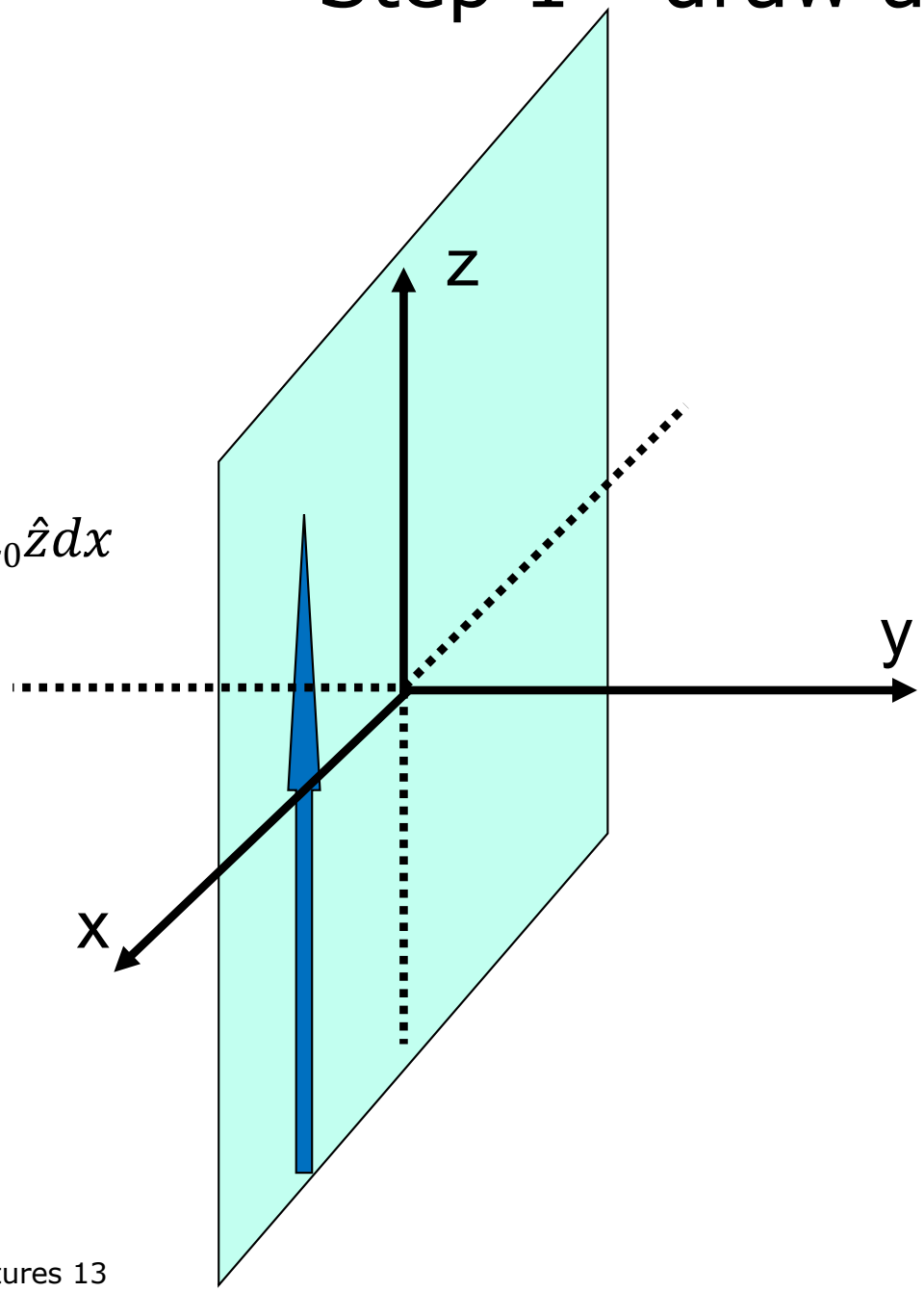
Surface current density $\vec{J}_s = \vec{J}_{s0}\hat{z}$ (A/m) flowing on $y = 0$ plane.

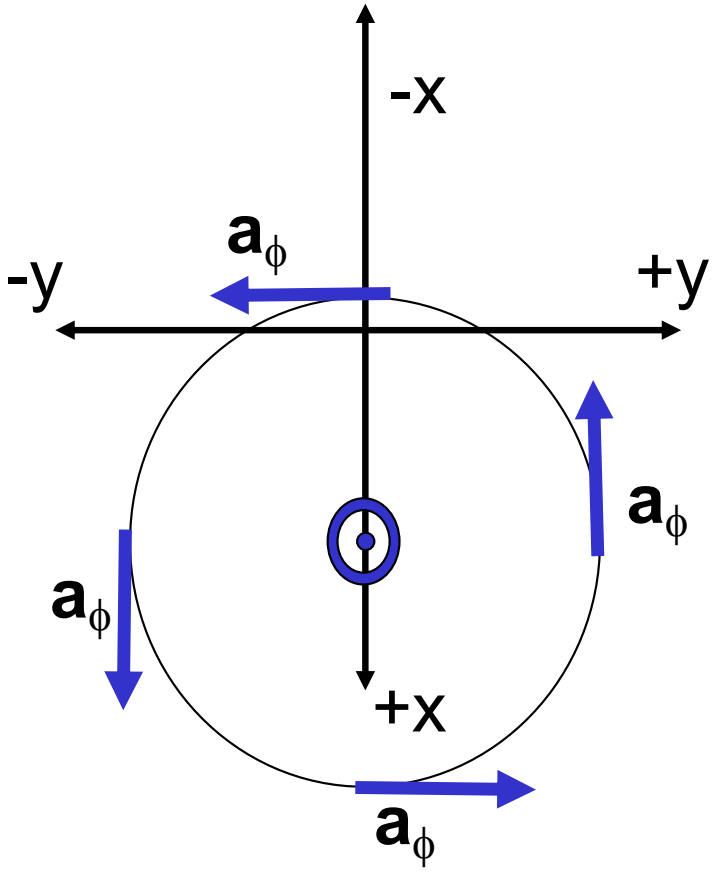
Solve for magnetic field $\vec{B}$ and $\vec{H}$



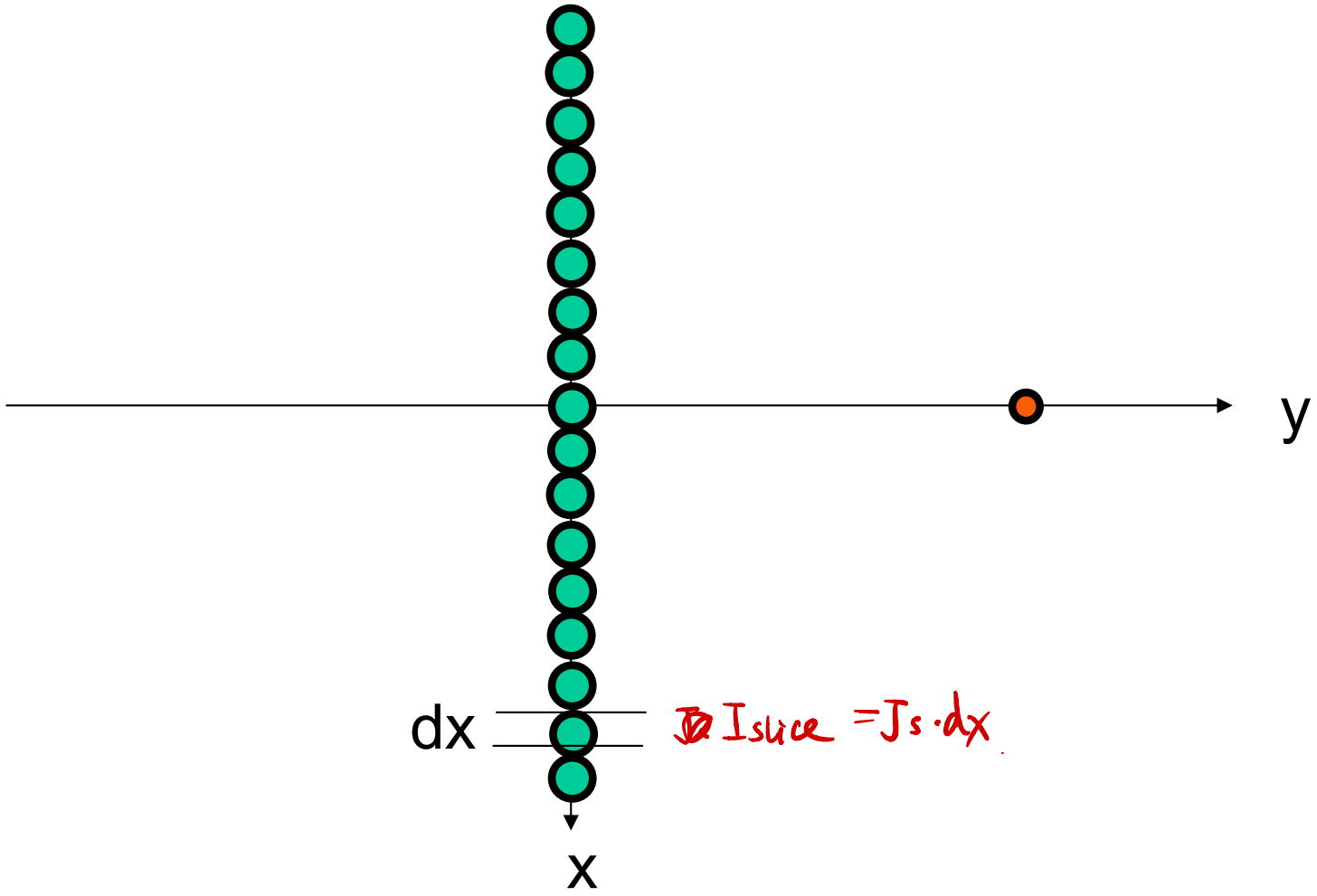
Step 1 - draw a picture

$$\boldsymbol{I}_s = \vec{\boldsymbol{J}}_{s0}\hat{z}dx$$

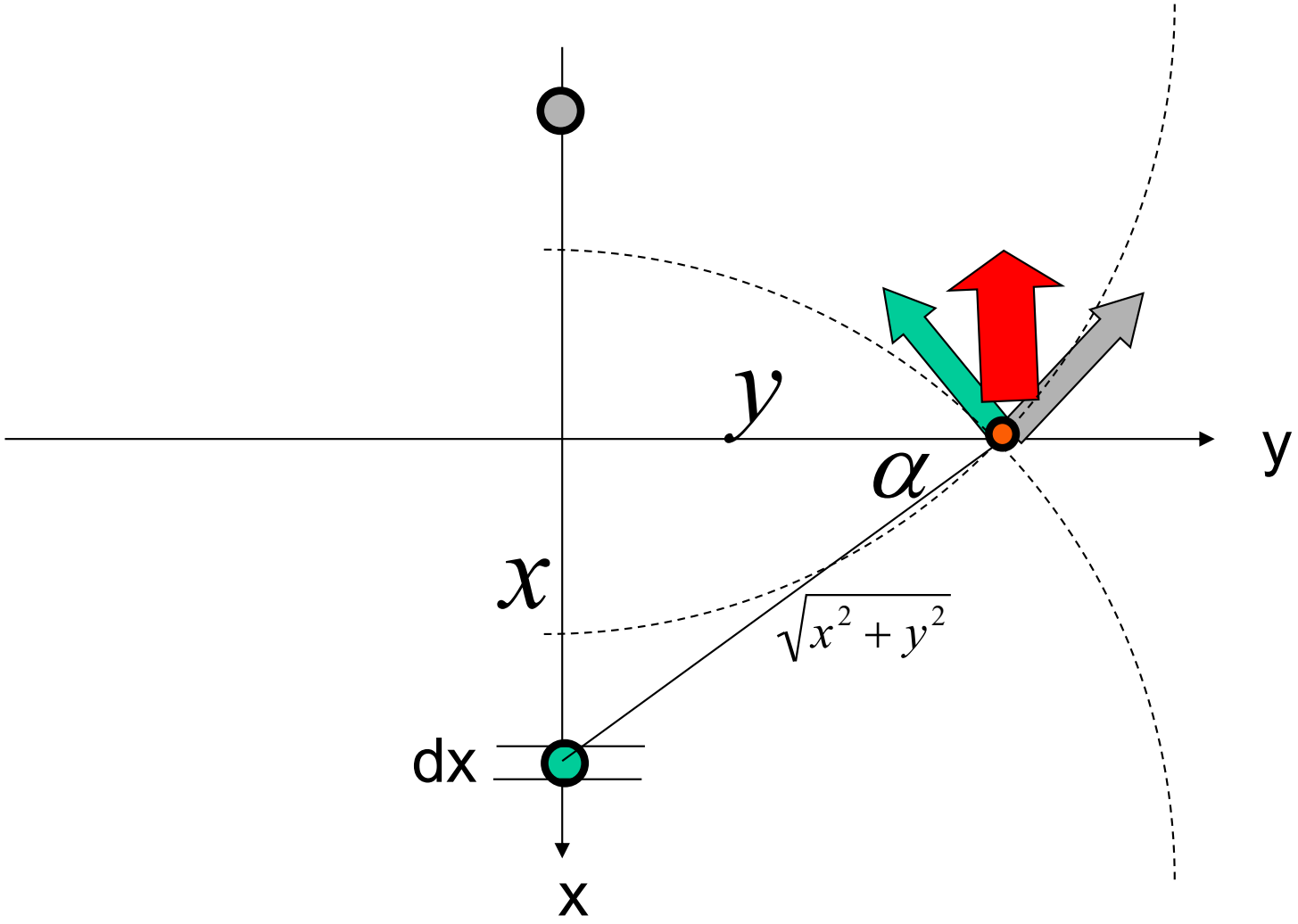




Step 2 - divide into segments

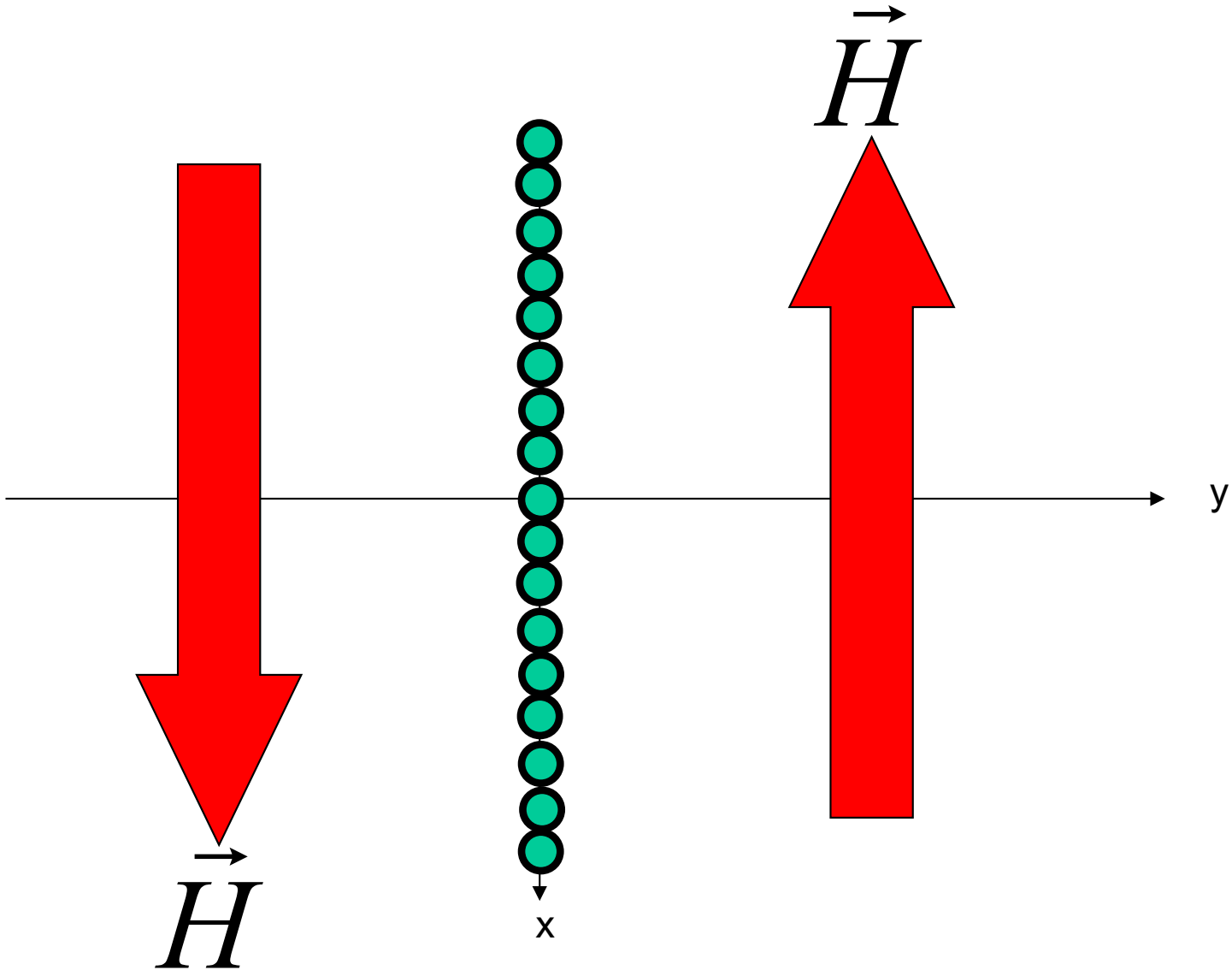


Step 3 - find d**H** for one small segment

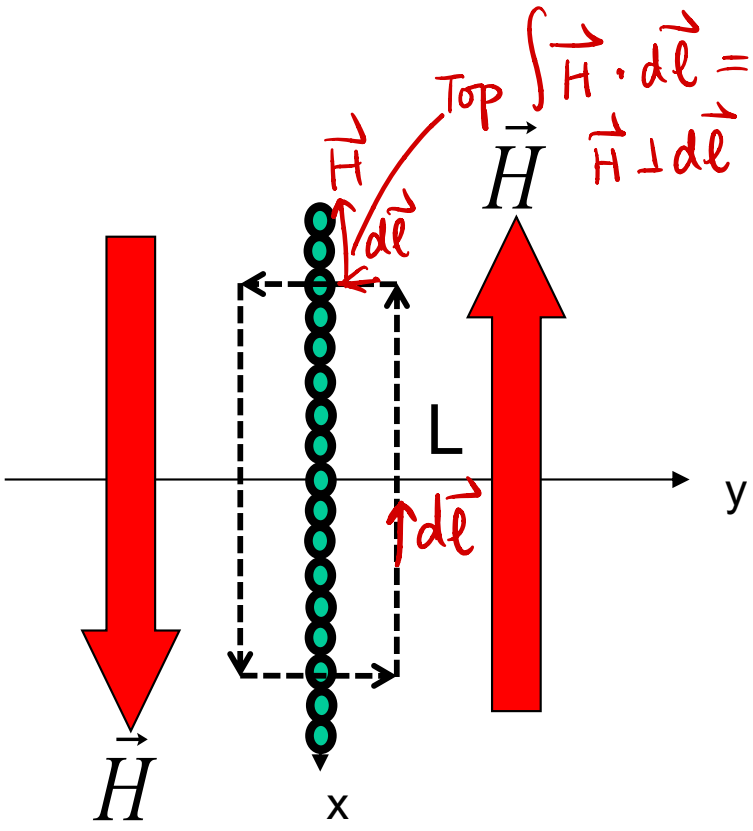


At every point (x,y), only the x-component remains, why?
For y>0, it is in $-\mathbf{a}_x$ direction

Step 4 – use symmetry to eliminate components that are zero



Aha! Now that we know H is along a_x, we can apply Ampere's Law to a simple path



$$\vec{J}_s = \vec{J}_{s0}\hat{z} \quad [A/m].$$

$$\oint_C \vec{H} \bullet d\vec{l} = I \quad [A].$$

Static Ampere's Law

$$\Rightarrow |H| \cdot L + |H| \cdot L = J_{s0} L$$

right. Left. [A].

$$|H| = \frac{1}{2} J_{s0}$$

$$\vec{\boldsymbol{H}} = -\frac{1}{2} J_{s0} \hat{x}\,\mathrm{sgn}(y)$$

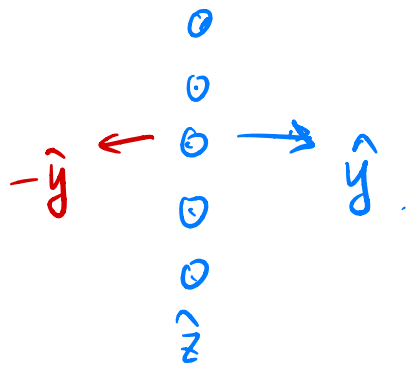
$$\vec{\boldsymbol{B}} = -\frac{\mu_0 J_{s0}}{2} \hat{x}\,\mathrm{sgn}(y)$$

General Formula for **H** due to infinite current sheet

The direction of current flow can be in ANY direction
The current sheet might not be on a coordinate plane

$+\hat{z} \times \hat{y}$ right
$-\hat{y}$ left.

$$\vec{H} = \frac{1}{2}\vec{J}_s \times \hat{a}_n$$

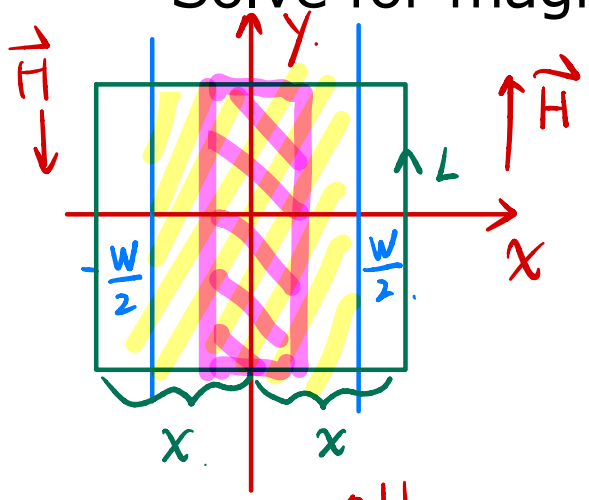


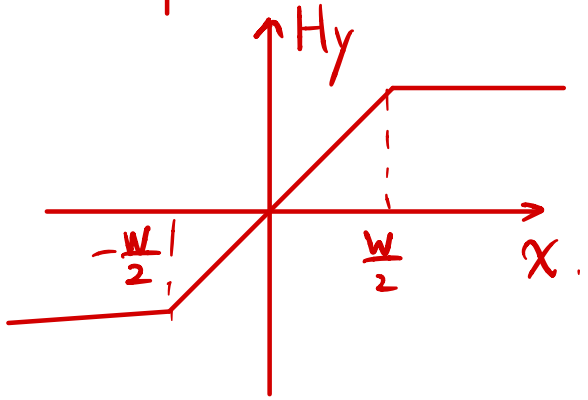
Unit vector normal to the surface

Example: Infinite Slab of Volumetric Current

Volumetric current density $\vec{J}$ (A/m²) flowing over $-\frac{W}{2} < x < \frac{W}{2}$.

Solve for magnetic field $\vec{B}$ and $\vec{H}$





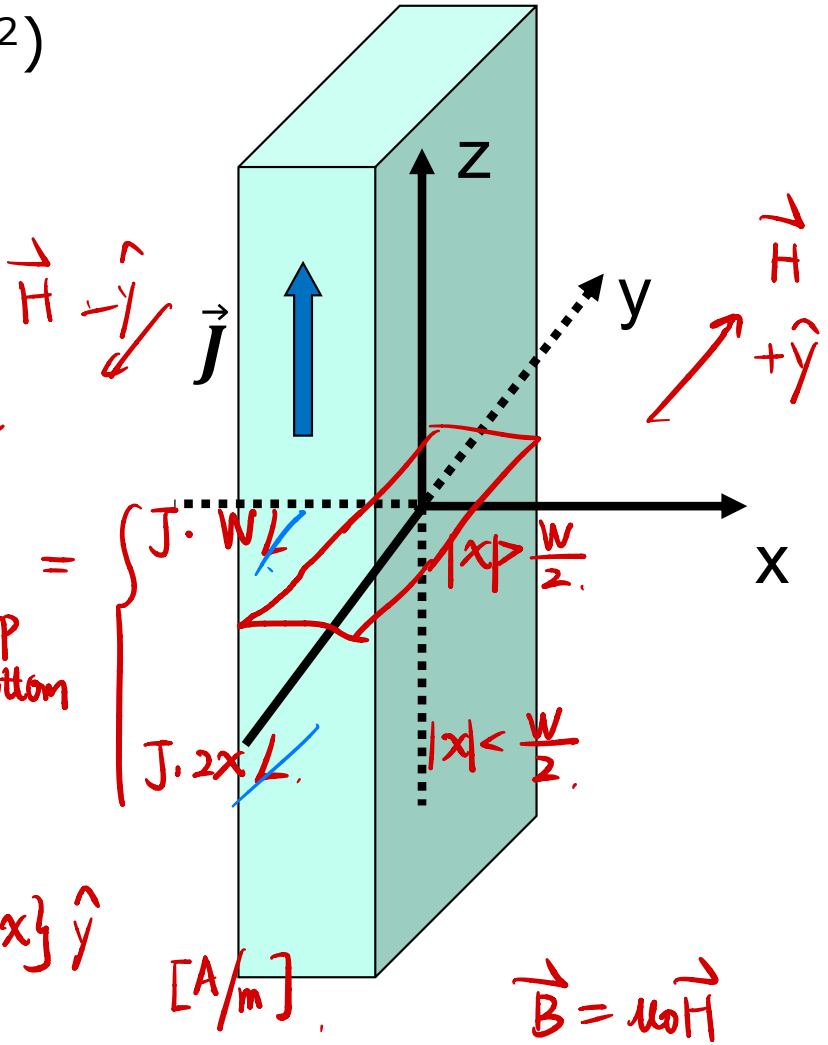
$$\oint_c \vec{H} \cdot d\vec{l} = I_{enc}$$

$$H \cdot L + H \cdot L + 0 = \begin{cases} J \cdot W L & |x| > \frac{W}{2} \\ J \cdot 2x L & |x| < \frac{W}{2} \end{cases}$$

right left top bottom

$$H_y = \begin{cases} J\frac{W}{2} \operatorname{sgn}\{x\}\,\hat{y} & [A/m] \\ Jx\,\hat{y} & \end{cases}$$

$$\vec{B} = \mu_0 \vec{H}$$



Example: Infinite solenoid

loops / per unit length.

Infinite solenoid with n loops per unit length [1/m] carrying I amps per loop. The magnetic field outside of the solenoid is 0.

Solve for magnetic field $\vec{B}$ and $\vec{H}$

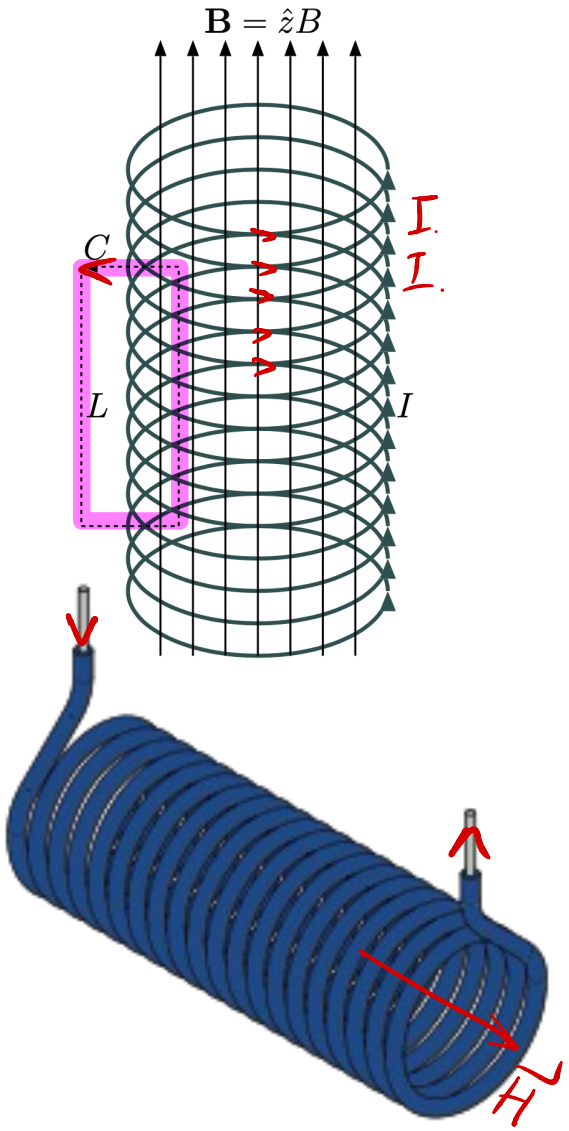
$$\oint_C \vec{H} \cdot d\vec{\ell} = I_{enc}$$

Inside.

$$H \cdot L = I \cdot n \cdot L$$

$$\vec{H} = nI\,\hat{z} \quad [A/m]$$

$$\vec{B} = \mu_0 \vec{H} = \mu_0 nI\,\hat{z} \quad [T]$$

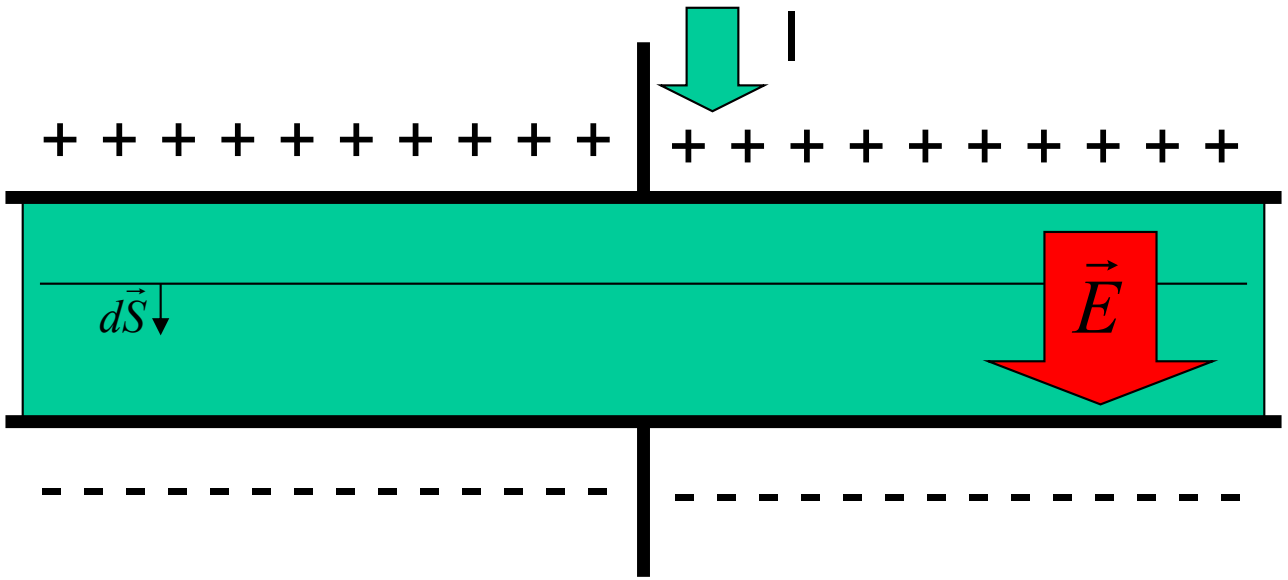


Problem with Static Ampere's Law

MMF $\oint_C \vec{H} \bullet d\vec{l} = I_{enclosed}$

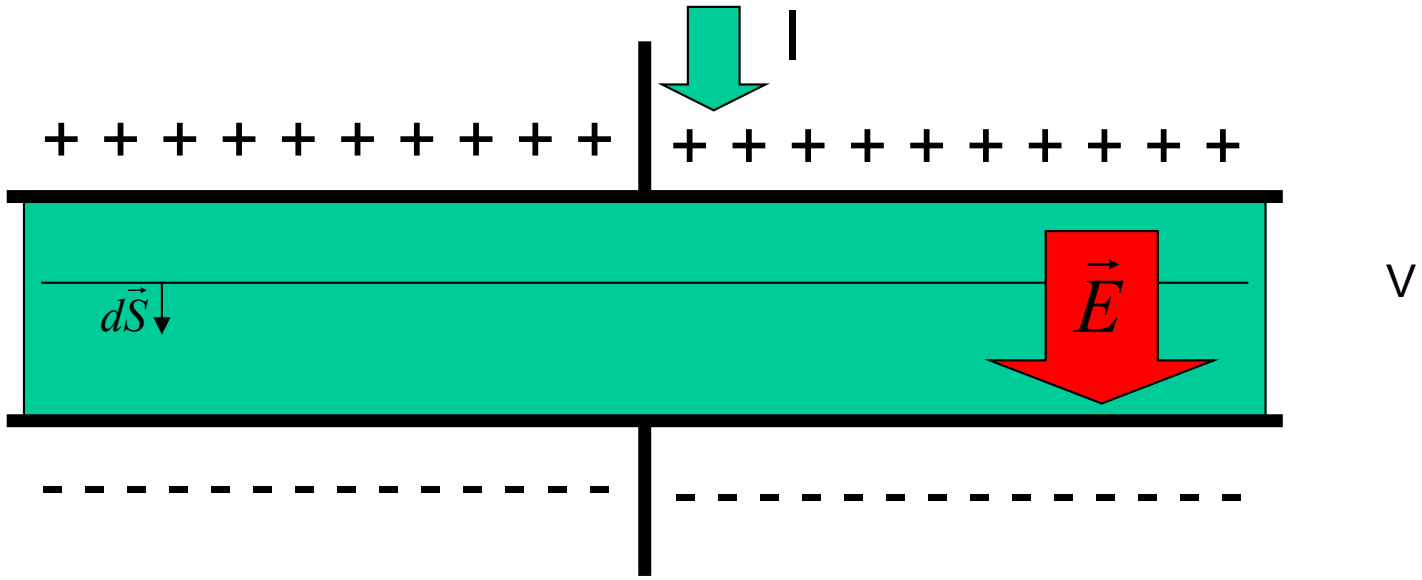
Consider a simple capacitor

- No DC current goes through
- But, AC voltage results in current flow



How does AC current get through a capacitor if it is not conducted through by a wire?

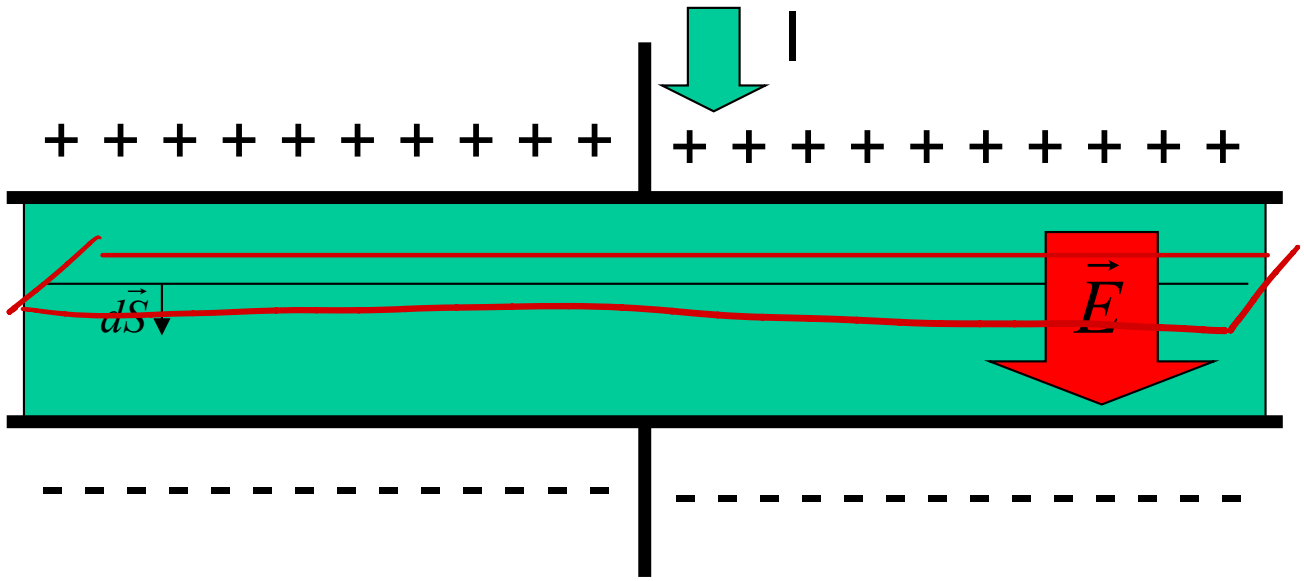
Problem with Static Ampere's Law



There is a second "way" to get current to flow

Somehow, a TIME-VARYING E field results in current flow

Problem with Static Ampere's Law



Flux of E field lines crossing a surface, S

$$\psi_E = \iint_S \varepsilon_0 \vec{E} \bullet d\vec{S}$$

Recall the Definition:
Electric Flux ψ_E Units: (C)

Gauss's Law

$$\oint\oint_S \vec{D} \cdot d\vec{s} = Q$$

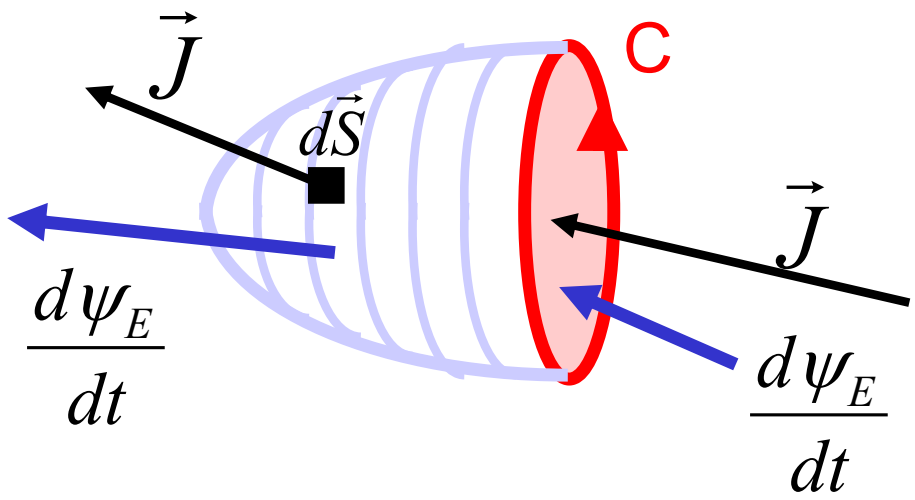
$[C/m^2] \cdot [m^2]$

James Clerk Maxwell's Genius Breakthrough

There are TWO sources of MMF:

1. Flow of charges due to current
2. Time-varying electric field

Called (by Maxwell)
"Displacement Current"



I_C

$$\oint_C \vec{H} \bullet d\vec{l} = \iint_S \vec{J} \bullet d\vec{S} + \frac{d}{dt} \iint_S \vec{D} \bullet d\vec{S}$$

MMF (Amps) = "Regular" Current (Amps) + Displacement Current (Amps)

"New" Definition: Displacement Flux Density Vector

$\vec{E}$ "Electric Field Intensity Vector" V/m

$\vec{D} = \varepsilon_0 \vec{E}$ "Displacement Flux Density Vector"

Units: $\vec{D} = \dfrac{\text{Charge}}{\text{Area}} = \dfrac{C}{m^2}$

$$\psi_E = \iint_S \varepsilon_0 \vec{E} \bullet d\vec{S} = \iint_S \vec{D} \bullet d\vec{S}$$ Coul

$$I_d = \frac{d\psi_E}{dt} = \frac{d}{dt} \iint_S \vec{D} \bullet d\vec{S}$$ Coul/sec (Amps)

Displacement current from a time varying E-field

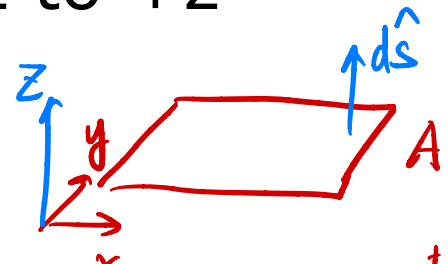
- Displacement current

$$I_d = \frac{d}{dt}\iint_S \vec{D} \bullet d\vec{S}$$

- Find the displacement current crossing an area $A=0.1m^2$ in the xy plane from the −z to +z side for:

$$\mathbf{E} = E_0\, t\, e^{-t^2}\mathbf{a}_z$$

const.



$$\psi_E = \iint_S \vec{D}\cdot d\vec{s} = \iint_S \varepsilon_0 E_0 t\, e^{-t^2}\hat{z}\cdot dxdy\hat{z} = \varepsilon_0 A E_0 t e^{-t^2}\ [C]$$

$$I_d = \frac{d\psi_E}{dt} = \varepsilon_0 A E_0 \left(e^{-t^2} - te^{-t^2}\cdot 2t\right)\ [A].$$

From Discussion 2.8 (p 106) of old book

Ampere's Law (Non static)

The Third Maxwell Equation

$$\oint_S \nabla \times \vec{H} \cdot d\vec{S} = \oint_C \vec{H} \bullet d\vec{l} = \iint_S \vec{J} \bullet d\vec{S} + \frac{d}{dt} \iint_S \vec{D} \bullet d\vec{S}$$

MMF (Amps) = "Regular" Current (Amps) + Displacement Current (Amps)

$$\nabla \times \vec{H} = \vec{J} + \frac{d\vec{D}}{dt}$$

After using Stokes' theorem to convert to differential form

Lecture 13 Summary

- Ampere's Circuit Law (Static)
- Ampere's Circuit Law (Time varying)
- Examples
  - infinite line of current
  - infinite surface current
  - infinite slab of volume current
  - infinite solenoid
- Next class
  - Faraday's Law